DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

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Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

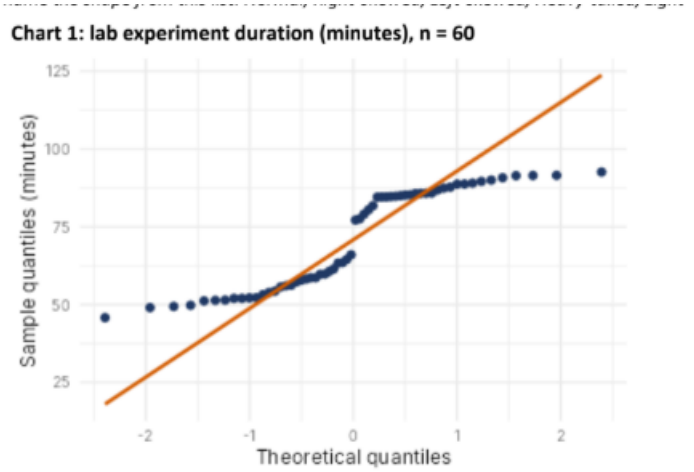


Chart 1: Lab Experiment Duration (n = 60)

Normal? No

Shape: S-shaped

Evidence in the plot:
The points do not follow the red reference line. Instead, they form a clear **S-shaped pattern**. This indicates that the data has a systematic departure from normality.

Interpretation:
The lab experiment duration data is **not normally distributed**. The S-shaped pattern suggests that the distribution has a different shape from a normal distribution.

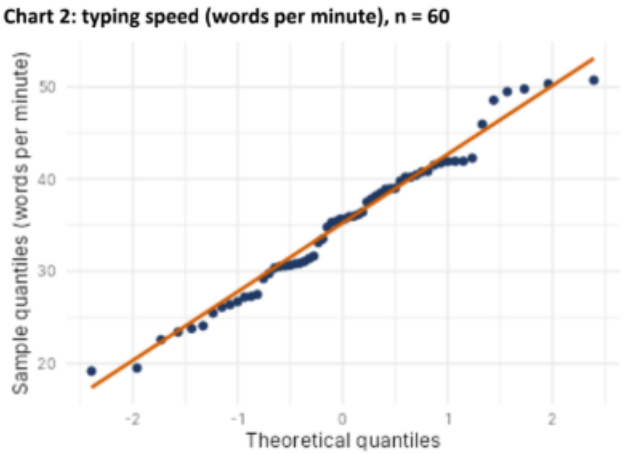


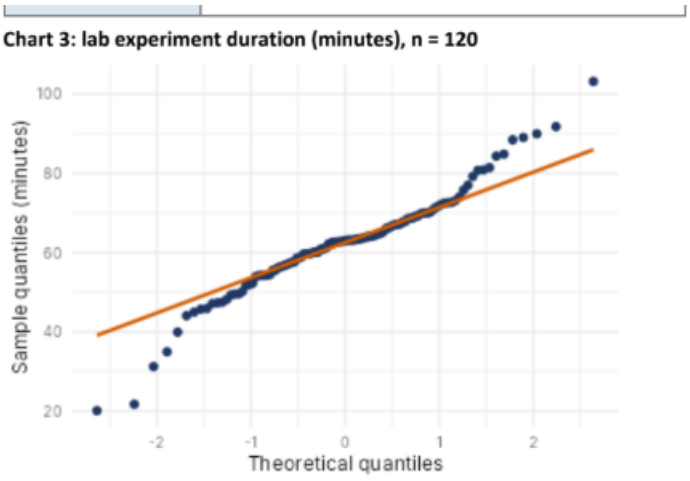
Chart 2: Typing Speed (n = 60)

Normal? Yes

Shape: Approximately normal

Evidence in the plot:
Most of the points are positioned **close to the red reference line**. There are only small deviations at some points, which are acceptable.

Interpretation:
The typing speed data is **approximately normally distributed**. Therefore, normality-based statistical methods can generally be considered appropriate.



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: Lab Experiment Duration (n = 120)

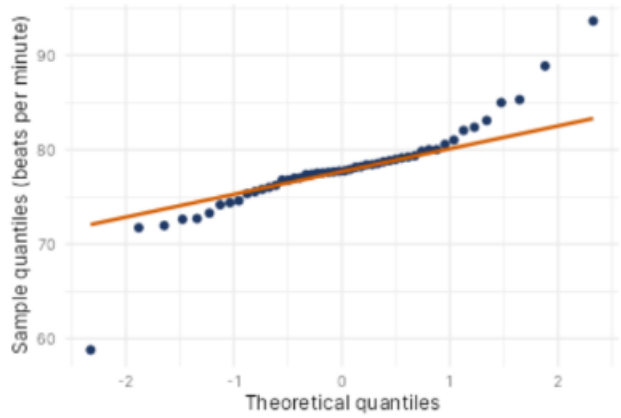
Normal? No

Shape: Right-skewed

Evidence in the plot:
The points deviate from the red reference line, particularly toward the **upper/right side**. The right tail extends noticeably above the reference line.

Interpretation:
The lab experiment duration data is **not normally distributed** and shows **right skewness**. This means that some larger values extend the distribution toward the right.

Chart 4: resting pulse (beats per minute), n = 50



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: Typing Speed (n = 100)

Normal? No

Shape: Left-skewed

Evidence in the plot:
The points show noticeable deviation from the red reference line, especially at the **lower/left side** of the plot.

Interpretation:
The typing speed data is **not normally distributed** and shows **left skewness**. The lower values create a longer or more pronounced tail on the left side.

Part B (4 marks)

B1. Chart 2 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Part B

B1. Chart 4 – Departure and Remedy

Chart 4 shows a **left-skewed distribution**. Because the data is not normally distributed, directly using methods based on the mean and standard deviation may not be appropriate.

A **square transformation** can be tried to reduce the skewness and make the distribution closer to normal.

Answer:

Departure: Left-skewed distribution.
Remedy: Try a square transformation before using mean-and-SD methods. Other possible approaches include a suitable transformation or robust statistical methods.

B2. R Code for Q-Q Plot

The question gives:

ggplot(df, aes(sample = value)) + _____ + _____

The two missing functions are:

stat_qq()

stat_qq_line()

Complete R code

```
library(ggplot2)


ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```